Unit: 6 Unit 6: Systems of Linear Equations and Inequalities **Name:** ____________
Topic: Linear Inequalities with Two Variables **Date:** __________

Foundation (Jalapeno)

Q1. What is the graph of the inequality $x + y > 2$?
(A) Solid line with shading above
(B) Dashed line with shading below
(C) Solid line with shading below
(D) Dashed line with shading above
(E) No shading

Q2. Which inequality has a solid boundary line?
(A) $x + ygeq2$
(B) $x + y > 2$
(C) $x + yleq2$
(D) $x + y < 2$
(E) $x - y = 2$

Q3. What is the solution region for $x > 2$ and $y < 3$?
(A) Quadrant I
(B) Quadrant II
(C) Quadrant III
(D) Quadrant IV
(E) All quadrants

Q4. The inequality $y < x$ has a
(A) Horizontal dashed line
(B) Vertical dashed line
(C) Diagonal dashed line
(D) Horizontal solid line
(E) Vertical solid line

Q5. For the inequality $x + yleq4$, what is the boundary line?
(A) Dashed line
(B) Solid line
(C) No line
(D) Vertical line
(E) Horizontal line

Q6. The inequality $y > x - 2$ has a
(A) Diagonal dashed line with shading above
(B) Diagonal solid line with shading below
(C) Vertical dashed line with shading to the right
(D) Horizontal solid line with shading above
(E) Horizontal dashed line with shading below

Q7. Which of the following inequalities has a dashed boundary line?
(A) $x + ygeq3$
(B) $x + yleq3$
(C) $x + y > 3$
(D) $x + y < 3$
(E) $x + y = 3$

Q8. The inequality $ygeq - x$ has a
(A) Diagonal solid line with shading above
(B) Diagonal dashed line with shading below
(C) Vertical solid line with shading to the right
(D) Horizontal dashed line with shading above
(E) Horizontal solid line with shading below

Open Ended Questions (FRQ)

Q9. Graph the inequality $x + 2y > 4$. Identify the boundary line and the solution region.

Q10. Solve the system of inequalities: $x + yleq3$ and $x - y > 1$. Show the solution region and the boundary lines.

Chef's Tips

- Ensure students have graph paper for questions 9 and 10
- Check for correct identification of solid and dashed boundary lines
- Remind students to shade the correct region for each inequality